

Table S1: Population QC summary over the 801-subject primary cohort (frozen `table_03_qc_summary.csv`; mean and median across subjects, primary configuration). Strength is in SIFT-weighted streamline-count units.

Metric	Mean	Median
Nodes per graph	456.0	456.0
Raw density	0.4515	0.4623
Isolated nodes after threshold (mean)	0.1200	0.0000
Multi-node components (mean)	1.0000	1.0000
Giant-component fraction	0.9997	1.0000
Global efficiency E_0	0.5467	0.5468
Mean node strength	15,588.4	15,527.4
<code>flag_density_iqr</code>	0.0000	0.0000
<code>flag_strength_iqr</code>	0.0056	0.0000
<code>flag_components</code>	0.0000	0.0000
<code>flag_isolated</code>	0.1044	0.0000
<code>n_qc_pass</code>	801.0	801.0

Supplementary Material for **“Control-impact architecture of the human structural** **connectome: degree dominance, a small near-universal** **residual, and a cross-scale architectural comparison with the** **fly connectome”**

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S1 Dataset provenance

Zenodo 19796783 (AOMIC-ID1000 structural connectomes), CC-BY-4.0; 10 zips, 7,067,175,601 bytes, byte-exact vs. API; CRC clean; 900 subjects $\times$ 7 atlases $\times$ 4 variants. Verification: `../00_Metadata/AOMIC_VERIFICATION/checksums: ../00_Metadata/CHECKSUMS_AOMIC.sha256`. Raw zips are read-forever (never modified).

S2 QC rules

Pre-registered property-based QC (E02, CIS-blind): dimension mismatch, NaN, Inf, asymmetry $> 10^{-9}$, negative off-diagonals, isolated nodes, strength IQR, density IQR, multi-component (> 1 non-trivial component). Cohort: 801/900 pass; 94 isolated-node + 5 strength-IQR flags; flagged subjects retained, never deleted. Population QC summary: Table S1; flag distributions: Fig. S1. Artifacts: `03_BASELINE/qc_primary.csv`, `BASELINE_RESULTS.md`, `01_RAW_PROBES/raw_qc_report.csv`.

S3 CIS mathematical definition

$$\text{CIS}(i) = \frac{E(G) - E(G - i)}{E(G)}; \quad E(G) = \frac{1}{N(N-1)} \sum_{\substack{u,v=1 \\ u \neq v}}^N \frac{1}{d(u,v)} \quad (1)$$

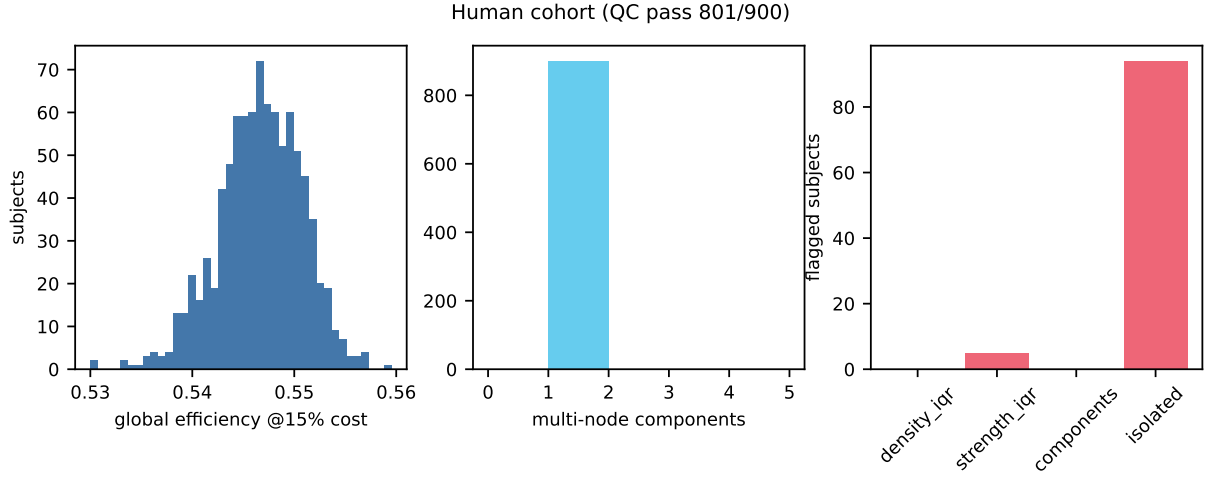


Figure S1: Population QC (900 subjects). QC flag distributions for the full acquired cohort; 99 subjects carry flags (94 isolated-node, 5 strength-IQR), retained rather than deleted; 801 subjects pass. Cohort selection was frozen before any CIS computation.

on the binary undirected thresholded graph; $E(G-i)$ on the induced $(N-1)$ -node subgraph $((N-1)(N-2)$ normalization). Exact computation (scipy Dijkstra); reference-vs-naive validation 6.1×10^{-16} (04_CIS/REFERENCE_IMPLEMENTATION_VALIDATION.md); boolean-matmul fast path used only for null graphs, equivalence-gated at 10^{-9} (max dev 4×10^{-16}). Implementation: 02_PREPROCESSING/cache_io.py (single source of truth).

S4 Null-model algorithm

Undirected Maslov–Sneppen double-edge swaps: canonicalized edge list (deduplicated symmetric orientations), $n_{\text{swaps}} = 10 |E|$ attempts per null, rejecting self-loops, duplicates, degenerate picks; exact per-node degree verification per null (10,000/10,000 exact; 0 self-loops); rejection redraw (max 5 attempts). Adversarial regression suite (hub-selfloop/pendant, dense $p = 0.6$, ring): regression_tests_undirected.csv/.json.

S5 Degree/strength matching

$\pm 10\%$ total degree, 1:1 greedy, no replacement; nearest-50 fallback flagged per pair. 36,846 degree-matched node pairs (top-decile CIS nodes vs matched controls). Residual defined as observed minus matched-control CIS; strength association reported alongside (median $\rho(\text{CIS}, \text{strength}) = 0.486$). Artifacts: 04_CIS/degree_control.py, degree_strength_control_summary.json, DEGREE_STRENGTH_CONTROL.md.

S6 Threshold parameters

$k = \text{round}(c \cdot N(N-1)/2)$ largest $|\cdot|$ off-diagonal weights; diagonal zeroed; binary output. Primary $c = 0.15$ ($k = 15,561$); ladder $\{0.10, 0.20, 0.25\}$. Rationale: raw densities 0.45–0.88 preclude unthresholded path-length statistics (STUDY_DESIGN §3.2). GE ladder at 4S456: 0.4911/0.5468/0.5877/0.6203.

S7 Statistical tests

Battery B: per-subject z vs. null distribution of top-50 median CIS; population two-sided sign test (binomial). Battery A: per-node empirical one-sided p with add-one correction ($p = (\#\{\text{nulls} \geq \text{obs}\} +$

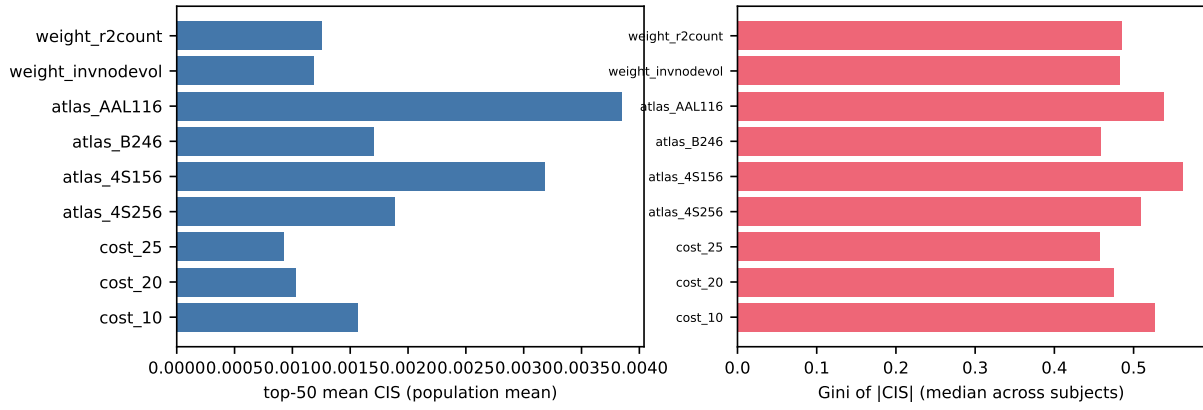


Figure S2: Robustness ladder (9 configurations, $n = 150$ per configuration). Top-50 mean-of-means and Gini across atlases {4S256, 4S156, Brainnetome246Ext, AAL116}, weights {sift_invnodevol, radius2_count}, and costs {0.10, 0.20, 0.25}. All 9 configurations reproduce the primary architecture; the maximum like-for-like top-50-mean deviation is 0.00262 (atlas AAL116; no numeric tolerance was pre-registered for E06 — descriptive comparison).

$1)/(100+1)) \rightarrow$ Stouffer $z (\sum z/\sqrt{n}) \rightarrow$ BH-FDR $q < .05$ (single stratum). Enrichment: control A = node-label shuffle (10,000 perms); control B = degree-matched peer pools resampled 10,000 $\times$; z and empirical p with add-one floor. Rank stability: 100 seeded half-splits, Spearman of mean-CIS ranks.

S8 FDR procedure

Benjamini–Hochberg over 456 node p -values (single stratum = primary configuration), $q = 0.05$; 43 survivors; censored nodes (all-null $\geq$ observed; 413 nodes) carry $p = 1$ and Stouffer $z = -\infty$ by construction.

S9 Robustness configurations

9 conditions on frozen $n = 150$ cohort: atlases {4S256, 4S156, Brainnetome246Ext, AAL116}, weights {sift_invnodevol, radius2_count}, costs {0.10, 0.20, 0.25}. All 9 conditions reproduce the primary architecture; max like-for-like top-50-mean deviation 0.00262 (atlas AAL116; full precision 0.0026194, primary = 0.0012286 on the same $n = 150$ cohort — RESOLUTION.md concordance). Note: no numeric tolerance was pre-registered for E06; the concordance is a descriptive comparison, not a pass/fail gate. Artifacts: 06_ROBUSTNESS/, 09_TABLES/table_08_robustness.csv. Ladder: Fig. S2.

S10 Seeds

Subject samples: one RNG, seed 20260922, nested samples ($A \subset B$). Nulls: 100+ i per graph. Enrichment/rank-stability: 20260922. All recorded in null_manifest_battery_a.csv and script constants.

S11 Additional distributions

Null δ/z distributions: battery_a/*.npz (100 $\times$ 456 null-CIS matrices per subject), null_results_battery_b.csv (per-subject null mean/sd). Per-subject CIS: 04_CIS/subject_cis/sub- $\langle$ subjectID $\rangle$.csv (456 rows each, 900 files). Baseline: 09_TABLES/table_baseline_network_properties.csv. Highest-stability nodes: Table S2.

Table S2: Eight highest-ranked nodes by population-mean CIS (frozen table_05_top_stability.csv): mean CIS across the 801-subject cohort, mean degree, fraction of subjects in which the node falls in the top-50 (top50_rate) and top decile (topdecile_rate) CIS. Node indices follow the frozen 4S456 atlas manifest.

Rank	Node	mean CIS	mean degree	top-50 rate	top-decile rate
1	415	0.0037	226.9	1.000	1.000
2	401	0.0034	213.8	1.000	1.000
3	414	0.0046	235.9	1.000	1.000
4	400	0.0043	226.9	1.000	1.000
5	442	0.0020	159.1	0.993	0.992
6	451	0.0020	143.5	0.987	0.979
7	444	0.0018	157.5	0.982	0.976
8	97	0.0012	160.6	0.968	0.958

S12 Anatomical mapping

00_MANIFEST/manifests/atlas_4S456_system_labels.csv (456 nodes; Vis 61, SomMot 77, DorsAttn 46, SalVentAttn 47, Limbic 26, Cont 52, Default 91, Subcortical_Cerebellar 56). Human chokepoint catalogue: 07_CROSS_SCALE/human_chokepoint_catalogue.csv.

S13 Literature novelty matrix

10_LITERATURE/NOVELTY_MATRIX.md (+ Kudriavtsev gate record: 10_LITERATURE/KUDRIAVTSEV_2026_FULLTEXT_REVIEW.md).

S14 Reproducibility instructions

1. `git clone` the repository (heavy data excluded; raw zips from Zenodo).
2. Run 02_PREPROCESSING/preprocess_v2.py (rebuilds cache_parts/, resume-safe) → 03_BASELINE/baseline.py → 04_CIS/cis_primary.py → 05_NULLS/degree_preserving/null_models.py --battery alb → 04_CIS/e05_statistics.py → 06_ROBUSTNESS/robustness.py → 09_TABLES/build_tables.py → 08_FIGURES/make_figures.py.
3. Numeric audit without any recomputation: `python 10_REPORT/verify_final_numbers.py` (stdlib-only; 46 checks).
4. Environment: Python 3.13, numpy 2.4.2, scipy 1.18.0, pandas 2.3.3 (see ANALYSIS_FREEZE.md provenance-gap note).

S15 Conceptual schematic

Two-layer architecture: a dominant connectivity-dependent component plus a smaller degree-independent residual, arbitrated by matched controls and degree-preserving nulls. Schematic only; no data are plotted (Fig. S3).

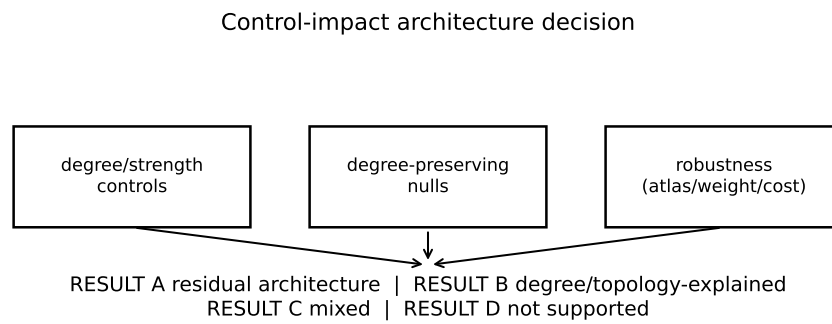


Figure S3: Conceptual schematic. Two-layer architecture: a dominant connectivity-dependent component plus a smaller degree-independent residual, arbitrated by matched controls and degree-preserving nulls. Schematic only; no data are plotted.